សាកលវិទ្យាល័យ បៀលធី អន្តរជាតិ

**BELTEI INTERNATIONAL UNIVERSITY**

គុណភាព ប្រសិទ្ធភាព ឧត្តមភាព សីលធម៌ គុណធម៌ Quality Efficiency Excellence Morality Virtue

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# មហាវិទ្យាល័យ ព័ត៌មានវិទ្យា និង វិទ្យាសាស្ត្រ

**FACULTY OF INFORMATION TECHNOLOGY AND SCIENCE**

**MAJOR: SOFTWARE ENGINEERING**

**SUBJECT: SOFTWARE PROJECT MANAGEMENT**

**TOPIC: “AI TRAVEL GUIDE CHATBOT FOR CAMBODIA”**

**FINAL PROJECT**

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# TOPIC: “AI TRAVEL GUIDE CHATBOT FOR CAMBODIA”

# 1. PROJECT PROPOSAL

## 1.1. Introduction

This project is an AI-powered travel guide chatbot that assists domestic and international tourists with travel information by:

- Recommending places in Cambodia (Angkor Wat, Phnom Penh, Siem Reap, beaches, cultural heritage sites, and natural attractions, etc.).
- Giving travel tips (transport, food, culture, weather, safety)
- Answer tourists' questions using English (and optionally Khmer)
- Provide 24/7 instant responses to users
- Improve tourist experience and satisfaction
- Accessible via web or mobile application
- Easy-to-use and user-friendly interface s
- Promote Cambodia’s tourism industry using modern AI technology

To address these problems, this project proposes the development of an AI Travel Guide Chatbot for Cambodia that provides instant, accurate, and personalized travel information using Artificial Intelligence.

## 1.2. Objectives

The main objectives of the AI Travel Guide Chatbot for Cambodia project are designed to address the problems faced by tourists and improve their overall travel experience:

- Build a chatbot to assist tourists
  Develop an AI-powered chatbot that can interact with users in natural language, answer travel-related questions, and provide guidance throughout their trip.
- Provide travel info and recommendations
  Deliver accurate and up-to-date information about tourist destinations, local attractions, travel tips, cultural practices, transportation options, weather, and safety. Offer **personalized recommendations** based on user preferences.
- Improve tourist experience in Cambodia
  Enhance the convenience, efficiency, and satisfaction of tourists by providing **24/7 instant support**, overcoming language barriers, and creating an engaging and user-friendly interface. Contribute to the promotion of Cambodia’s tourism industry through modern AI technology.

## 1.3. Scope of the Project

**Includes:**

- Development of an AI-powered chatbot that provides travel information about Cambodia.
- Information on tourist attractions, historical sites, natural landmarks, cultural experiences, beaches, and local events.
- Travel tips including transportation, food, weather, safety, and cultural practices.
- Personalized recommendations based on user preferences.
- Support for English language interaction (optionally Khmer).
- Availability through web and mobile applications with a responsive, user-friendly interface.
- 24/7 instant response capability using AI and NLP (Natural Language Processing).
- Collection of user feedback for future improvements of the chatbot.

**Excludes:**

- Booking or reservation services for hotels, transport, or tours.
- Payment processing or financial transactions.
- Detailed travel information outside of Cambodia.
- Human customer support (the system is fully automated).
- Offline functionality; the chatbot requires an internet connection.
- Personalized medical, legal, or emergency advice beyond general safety tips.

## 1.4. Methodology

This project will adopt the **Agile Scrum methodology** to ensure flexibility, continuous improvement, and timely delivery. The development process will be divided into **2-week sprints**, where each sprint focuses on delivering a set of functional features.

Key activities in the Scrum process include:

- **Sprint Planning:** Define sprint goals and select features from the product backlog.
- **Development:** Design, coding, and integration of chatbot features.
- **Daily Stand-up Meetings:** Short meetings to track progress and resolve issues.
- **Sprint Review:** Demonstrate completed features to stakeholders.

- **Sprint Retrospective:** Evaluate team performance and identify improvements for the next sprint.

Using Agile Scrum allows the team to quickly adapt to requirement changes, improve software quality through continuous testing, and deliver a working AI Travel Guide Chatbot incrementally.

**Diagram Explanation**

The diagram illustrates the **Agile Scrum development cycle** used in the *AI Travel Guide Chatbot for Cambodia* project.

1. **Product Backlog**
   This is a prioritized list of all system requirements and features, such as chatbot responses, language support, and admin management.
2. **Sprint Planning**
   The team selects a set of backlog items to be completed in the upcoming sprint. Sprint goals and tasks are clearly defined.
3. **Sprint (2 Weeks)**
   During the sprint, the team performs:
   - System design
   - Coding
   - Integration
   - Testing
     All activities focus on delivering a working feature of the chatbot.
4. **Sprint Review**
   At the end of the sprint, the completed features are demonstrated and evaluated to ensure they meet the requirements.
5. **Sprint Retrospective**
   The team discusses what went well, what problems occurred, and how to improve the next sprint.
6. **Increment Release**
   A working version of the system is produced and added to the existing product. Feedback is then used to update the Product Backlog.

This iterative cycle continues until the final AI Travel Guide Chatbot system is completed.

**Why This Methodology is Suitable**

- Supports **changing requirements**
- Ensures **continuous testing**

- Delivers **working software frequently**
- Reduces project risks

- **Process:** We will use the Scrum framework with 2-week sprints to iteratively build and test features.
- **Tools:**
  - *Version Control:* Git & GitHub
  - *IDE:* Visual Studio Code
- **Technology Stack:**
  - ***Backend****:* Python (Django Framework)
    - Python (Django Framework) Handles API requests, chatbot logic, and communication with the AI service.
  - ***Frontend****:* HTML5, CSS3, Bootstrap/Tailwind, JavaScript
    - Used to build a responsive and interactive user interface for the chatbot.
  - ***Database****:* **PostgreSQL**
  - ***AI Libraries:*** EasyOCR, NumPy, Python Re (Regex)
  - **AI / NLP Engine**
    - Processes user input and generates intelligent responses.

## 1.5. Expected Outcome

At the end of this project, the following outcomes are expected:

- A fully functional **AI Travel Guide Chatbot for Cambodia** that provides accurate and relevant travel information.
- A web-based chatbot system accessible 24/7 through modern web browsers.
- Support for **English and Khmer languages** to assist both local and international tourists.
- Reduced time spent by users searching for travel information by at least **40%**.
- An admin panel that allows administrators to easily manage tourist data, FAQs, and chatbot content.
- Well-documented project deliverables including Project Proposal, SRS, SDD, Test Plan, and User Manual.
- A clean, structured, and professional **GitHub repository** containing source code and documentation.
- Improved user experience and increased awareness of tourist attractions in Cambodia.

### 1.6. References

Here is a proper academic "References" section formatted for your AI Travel Guide Chatbot for Cambodia project.


- IEEE Computer Society, *IEEE Std 830-1998: Software Requirements Specification*, IEEE, 1998.

- Schwaber, K., & Sutherland, J., *The Scrum Guide*, Scrum.org, 2020.
- Pressman, R. S., & Maxim, B. R., *Software Engineering: A Practitioner's Approach*, 8th Edition, McGraw-Hill, 2015.
- Sommerville, I., *Software Engineering*, 10th Edition, Pearson Education, 2016.
- Ministry of Tourism Cambodia, *Tourism Statistics and Information*, Cambodia.
- OpenAI, *Chatbot and Artificial Intelligence Concepts*, OpenAI Documentation.
- ISO/IEC 25010, *Systems and Software Quality Requirements and Evaluation (SQuaRE)*, ISO, 2011.

# 2. Software Requirements Specification (SRS)

## 2.1. Purpose

The SRS defines a formal "contract" of exactly what the **AI Travel Guide Chatbot for Cambodia** software will do. It ensures that all stakeholders (developers, project manager, admin, and users) have a clear understanding of the system's functional and non-functional requirements. This document serves as the foundation for system design, development, and testing.

## 2.2. Functional Requirements

Functional requirements describe the specific behaviors or functions of the system. Examples for the AI Travel Guide Chatbot:

### 2.2.1 User Authentication & Profile

| ID | Requirement Description |
|---|---|
| FR1 | The system shall allow users to chat with the AI chatbot for travel information. |
| FR2 | The system shall provide recommendations for tourist attractions, hotels, food, and transportation. |
| FR3 | The system shall support multiple languages (English and Khmer). |
| FR4 | The system shall allow administrators to add, update, and delete tourism content and FAQs. |
| FR5 | The system shall allow users to search for information using keywords or natural language queries. |

### 2.2.2 Expense Management (CRUD)

The system shall provide an Expense Management module to help users manage travel-related expenses during their trip in Cambodia.

Functional Requirements:

The system shall allow users to create new expense records, including expense name, category, amount, date, and description.

- The system shall allow users to view a list of all recorded expenses for a selected trip.
- The system shall allow users to update existing expense records.
- The system shall allow users to delete expense records when they are no longer needed.
- The system shall support expense categorization such as accommodation, food, transportation, entrance fees, and shopping.
- The system shall automatically calculate the total expenses for a trip.
- The system shall store all expense data securely and associate it with the corresponding user profile.

**2.2.3 AI & Automated Entry (The "Smart" Features)**

The AI & Automated Entry module represents the core intelligence of the AI Travel Guide Chatbot for Cambodia.

Functional Requirements:

- The system shall allow users to ask travel-related questions using natural language.
- The system shall use AI and Natural Language Processing (NLP) to understand user intent and generate relevant responses.
- The system shall provide automated travel recommendations, including destinations, itineraries, food suggestions, and cultural tips.
- The system shall suggest popular tourist attractions in Cambodia based on user preferences, trip duration, and budget.
- The system shall support automated expense entry suggestions by recognizing expense-related messages (e.g., “I spent $10 on food”).
- The system shall continuously improve response accuracy through updated training data and feedback.
- The system shall provide multilingual support (English as default; Khmer as future enhancement).

### 2.2.4 Dashboard & Reporting

The Dashboard & Reporting module provides users with a summarized view of their travel information and chatbot interactions.

Functional Requirements:

- The system shall display a user dashboard showing trip overview details such as destinations, total expenses, and recent activities.
- The system shall present expense summaries categorized by type (food, transport, accommodation, etc.).
- The system shall generate visual reports such as charts or graphs to show spending patterns.
- The system shall allow users to view chat history with the AI chatbot.
- The system shall allow users to filter reports by date or trip.
- The system shall support exporting reports in common formats (e.g., PDF or CSV) as a future enhancement.

## 2.3. Non-Functional Requirements

Non-functional requirements define system properties such as performance, usability, and reliability:

| Category | Requirement |
|---|---|
| Performance | The system shall respond to user queries within 2 seconds. |
| Availability | The system shall be available 24/7 for users. |
| Security | Admin login shall require secure authentication. |
| Usability | The chatbot interface shall be intuitive and easy to navigate. |
| Scalability | The system shall handle at least 500 concurrent users. |

## 2.4. Use Case Diagram

**Use Cases:** illustrate how users interact with the system. Key actors include:

- **User (Tourist)**: Interacts with the chatbot to ask questions about places, hotels, or food.
- **Admin**: Manages content, updates FAQs, monitors system performance.

**Example Use Case: “Ask About Tourist Attractions”**

1. User opens the chatbot interface.
2. User types a question, e.g., “Best places to visit in Siem Reap.”
3. Chatbot interprets the question using AI/NLP engine.
4. System retrieves information from the database.
5. Chatbot displays the recommended places to the user.

*(We can create a simple **Use Case diagram** using tools like Lucidchart, Draw.io, or even hand-drawn for the report.)*

## 2.5. References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- IEEE Std 830-1998: *Software Requirements Specification*
- Example: Airline Database System SRS (for structure and formatting)

# 3. Project Plan (WBS & Schedule)

## 3.1. Work Breakdown Structure (WBS)

### 1.0 Planning Phase – AI Travel Guide Chatbot for Cambodia

### 1.1 Project Initiation

- 1.1.1 Identify project objectives and scope
- 1.1.2 Identify stakeholders (students, tourists, supervisors)
- 1.1.3 Define project deliverables
- 1.1.4 Obtain project approval

### 1.2 Requirement Analysis

- 1.2.1 Collect functional requirements
    - Travel place recommendations
    - Travel tips (culture, food, transport, safety)
    - English-based chatbot interaction

### 1.3 Feasibility Study

- 1.3.1 Technical feasibility (AI, chatbot platform, APIs)
- 1.3.2 Economic feasibility (cost, tools, hosting)
- 1.3.3 Operational feasibility (users, usability)
- 1.3.4 Schedule feasibility

### 1.4 Project Planning

- 1.4.1 Develop Work Breakdown Structure (WBS)
- 1.4.2 Create project schedule (timeline, milestones)
- 1.4.3 Assign roles and responsibilities
- 1.4.4 Estimate effort and resources

```
1.0 Planning Phase
│
├── 1.1 Project Initiation
│   ├── 1.1.1 Define project objectives
│   ├── 1.1.2 Identify stakeholders
│   ├── 1.1.3 Define project scope
│   └── 1.1.4 Project approval
│
├── 1.2 Requirement Analysis
│   ├── 1.2.1 Gather functional requirements
│   │   ├── Place recommendations
│   │   ├── Travel tips (food, culture, transport)
│   │   └── English chatbot interaction
│   ├── 1.2.2 Gather non-functional requirements
│   │   ├── Performance
│   │   ├── Security
│   │   └── Usability
│   ├── 1.2.3 Analyze system constraints
│   └── 1.2.4 Finalize requirements
│
├── 1.3 Feasibility Study
│   ├── 1.3.1 Technical feasibility
│   ├── 1.3.2 Economic feasibility
│   ├── 1.3.3 Operational feasibility
│   └── 1.3.4 Schedule feasibility
│
├── 1.4 Project Planning
│   ├── 1.4.1 Develop WBS
│   ├── 1.4.2 Create project schedule
│   ├── 1.4.3 Resource allocation
│   └── 1.4.4 Effort estimation
│
├── 1.5 Risk Management
│   ├── 1.5.1 Identify risks
│   ├── 1.5.2 Risk analysis
│   └── 1.5.3 Risk mitigation plan
│
└── 1.6 Documentation
    ├── 1.6.1 Project Proposal
    ├── 1.6.2 Software Requirements Specification (SRS)
    └── 1.6.3 Review and approval
```

## 2.0 Design Phase

### 2.1 Database Design

The database design describes how data is stored, organized, and related within the system. An Entity Relationship Diagram (ERD) is developed to model the relationships between the core entities:

- Users: Stores user account and authentication information.
- Expenses: Stores expense records such as amount, date, and description.

- Categories: Defines expense categories for classification.
- Receipts: Stores uploaded receipt images and OCR-extracted text.
- The ERD ensures data integrity and supports one-to-many relationships, such as one user having multiple expenses.

## 2.2 UI/UX Design

**Use of Figma in UI/UX Design**

Figma is used as the primary UI/UX design tool for this project to design, prototype, and validate the user interface before implementation. Figma is a cloud-based design platform that enables real-time collaboration, consistency in design, and rapid iteration, making it suitable for modern web and mobile application development.

**1. Wireframing and Layout Design**

In the early design phase, low-fidelity wireframes are created in Figma to define the structure and layout of key system pages. These wireframes focus on functionality and user flow rather than visual styling. The main wireframes include:

- **Dashboard Page**: Displays total expenses, categorized summaries, monthly spending charts, and quick insights.
- **Upload Page**: Provides a simple interface for users to upload receipt images for OCR processing.
- **Expense List Page**: Shows detailed expense records with search, filter, and sorting options.

Using Figma allows the design team to easily adjust layouts, align components, and ensure logical navigation between pages.

**2. High-Fidelity UI Design**

After wireframes are approved, high-fidelity designs are developed in Figma. These designs include colors, typography, icons, spacing, and visual hierarchy to create a clean and professional user interface. Figma's reusable **components** (buttons, forms, cards, navigation bars) are used to maintain design consistency across all pages.

**3. Responsive Design**

Figma is used to design responsive layouts for different screen sizes, including desktop, tablet, and mobile devices. Auto Layout and constraints features help ensure that UI elements resize and adapt correctly, providing a smooth and intuitive user experience across devices.

**4. Prototyping and User Flow**

Interactive prototypes are created in Figma to simulate real user interactions such as:

- Navigating from the dashboard to expense details
- Uploading receipts
- Filtering and viewing expense records

These prototypes help stakeholders and developers understand system behavior without writing code.

**5. Collaboration and Feedback**

Figma's real-time collaboration features allow designers, developers, and project supervisors to review the UI designs simultaneously. Comments and annotations are used to collect feedback, suggest improvements, and make design decisions efficiently.

**6. Developer Handoff**

Figma supports smooth handoff to developers by providing design specifications such as spacing, colors, fonts, and asset exports. This reduces misunderstandings during implementation and ensures that the final system closely matches the approved design.

## 2.3 System Architecture

The system architecture illustrates how system components interact. A flowchart is used to represent the process of data handling:

**Image Upload → OCR Processing → Data Extraction → Database Storage → User Display**

This architecture supports modular development and efficient data processing.

## 3.0 Development Phase

The Development Phase focuses on implementing the system according to the approved design. This phase includes backend development, AI logic integration, and frontend implementation.

3.1 Backend – Core Development

- Setup Django project and applications
- Implement user authentication and authorization
- Develop CRUD APIs for expense management

3.2 Backend – AI Logic Development

- Implement _clean_price_token function using regular expressions to extract price values
- Implement _smart_extract function using strict keyword logic
- Integrate EasyOCR for receipt text recognition

3.3 Frontend Development

- Build a responsive dashboard interface
- Create forms for manual expense entry
- Develop receipt upload interface for OCR processing

## 4.0 Testing Phase

The Testing Phase ensures the system functions correctly and meets user requirements.

4.1 Unit Testing

- Test regular expressions against different receipt formats

4.2 Integration Testing

- Test the complete workflow from receipt upload to automatic data filling

4.3 User Acceptance Testing (UAT)

- Verify that keywords such as "Total Amount" are detected accurately

4.4 Bug Fixing and Performance Tuning

- Fix identified defects
- Optimize system performance and accuracy

## 5.0 Deployment & Documentation

The Deployment and Documentation phase prepares the system for final delivery and presentation.

**5.1 Final Code Cleanup**

- Refactor code
- Remove unused files and improve readability

**5.2 User Manual**

- Provide step-by-step instructions on using OCR and voice features

### 5.3 Final Report and Defense Preparation

- Prepare final documentation
- Organize slides and system demonstration for project defense

## 3.2. Project Schedule (Gantt Chart)

The project schedule defines the timeline for developing the system and ensures that all project activities are completed within the allocated time. The schedule is organized based on the Work Breakdown Structure (WBS) and is presented using a Gantt Chart, which illustrates task durations, sequencing, and dependencies across all project phases.

The project is planned to be completed over a period of 14 weeks, covering the Planning, Design, Development, Testing, and Deployment phases.

### 3.2.1 Project Timeline Overview

| Phase | Activities | Duration |
|---|---|---|
| Planning Phase | Proposal, SRS, feasibility study | Week 1 – Week 2 |
| Design Phase | ERD, UI/UX wireframes, system architecture | Week 3 – Week 4 |
| Development Phase | Backend, AI logic, frontend | Week 5 – Week 9 |
| Testing Phase | Unit, integration, UAT, bug fixing | Week 10 – Week 12 |
| Deployment & Documentation | Final code, user manual, defense prep | Week 13 – Week 14 |

### 3.2.2 Detailed Gantt Chart Schedule

| Task ID | Task Name | Start Week | End Week |
|---|---|---|---|
| 1.0 | Planning Phase | 1 | 2 |
| 1.1 | Project Proposal | 1 | 1 |
| 1.2 | Software Requirements Specification (SRS) | 1 | 2 |
| 2.0 | Design Phase | 3 | 4 |
| 2.1 | Database Design (ERD) | 3 | 3 |
| 2.2 | UI/UX Design (Wireframes) | 3 | 4 |
| 2.3 | System Architecture Design | 4 | 4 |
| 3.0 | Development Phase | 5 | 9 |
| 3.1 | Backend Core Development | 5 | 6 |
| 3.2 | Backend AI Logic (OCR & Extraction) | 6 | 8 |
| 3.3 | Frontend Development | 7 | 9 |
| 4.0 | Testing Phase | 10 | 12 |
| 4.1 | Unit Testing | 10 | 10 |
| 4.2 | Integration Testing | 11 | 11 |
| 4.3 | User Acceptance Testing (UAT) | 11 | 12 |
| 4.4 | Bug Fixing & Performance Tuning | 12 | 12 |

| Task ID | Task Name | Start Week | End Week |
|---|---|---|---|
| 5.0 | Deployment & Documentation | 13 | 14 |
| 5.1 | Final Code Cleanup | 13 | 13 |
| 5.2 | User Manual Preparation | 13 | 14 |
| 5.3 | Final Report & Defense Preparation | 14 | 14 |

## 4. System Design Document (SDD)

The AI & OCR Processing Module is responsible for understanding user queries, extracting information from images, and generating accurate responses. This module integrates natural language processing (NLP) and Optical Character Recognition (OCR) to process text from receipt images or travel queries.

Components of the Module:

1. OCR Engine
   - Uses EasyOCR to extract text from uploaded images.
   - Supports multiple formats and ensures accurate extraction of textual information.
   - Example Use Case: Extract total amounts or keywords from receipts for expense tracking.
2. Text Cleaning & Parsing Functions
   - _clean_price_token: Uses regex to identify and clean numeric values (prices) from extracted text.
   - _smart_extract: Uses keyword-based logic to identify relevant information and categorize it correctly.
3. Natural Language Processing (NLP)
   - Understands user queries in English.
   - Identifies intents (e.g., “Recommend places in Siem Reap”) and entities (e.g., “beaches,” “Angkor Wat”).
   - Searches the database and returns relevant travel tips or place information.

4. Response Generation
   - Formats output from the AI and OCR modules into user-friendly messages.
   - Ensures correct display in the chat interface.

### 4.4.2 Application Security Design

Security is a critical aspect of the system, ensuring that user data and system components are protected from unauthorized access, tampering, or leakage. The Application Security Design includes:

1. **User Authentication & Authorization**
   - Secure login mechanism using **hashed passwords**.
   - Role-based access control to restrict administrative features.
2. **Data Protection**
   - Sensitive data (e.g., passwords, personal info) is encrypted.
   - Database access is restricted to authorized modules only.
3. **Secure Communication**
   - All client-server interactions are transmitted via **HTTPS** to prevent eavesdropping.
   - APIs include authentication tokens to prevent unauthorized requests.
4. **Input Validation**
   - Validates all user inputs to prevent injection attacks or malicious data submission.
   - OCR-extracted text is sanitized before storing in the database.
5. **Logging & Monitoring**
   - Tracks user activities and system events for auditing and anomaly detection.
   - Helps identify potential security breaches or system misuse.

# 5. Test Plan and Test Summary Report

The **AI Travel Guide Chatbot for Cambodia** was successfully designed and developed. The system provides **instant, accurate, and personalized travel information** to tourists. It supports **English and Khmer**, helping overcome language barriers. The chatbot enhances **tourist experience and satisfaction** through 24/7 AI assistance. The project demonstrates how **modern AI technology** can support and promote Cambodia's tourism industry.

## 5.1. Test Plan Strategy

Scope: Unit Testing (Python functions), System Testing (Full flow), Regression Testing.
Methodology: Black Box Testing and Boundary Value Analysis.
Tools: Django Test Client, Chrome/Edge, Sample receipt images.

**Frontend:** Web browser (desktop and mobile), responsive layout testing.
**Backend:** Django server, AI modules, OCR engine (EasyOCR).
**Database:** Relational database with sample travel and user data.
**Tools:** Automated scripts for unit and integration testing, manual testing for UAT.

## 5.2. Test Cases (Execution Log)

The Test Cases Execution Log documents the tests performed on the AI Travel Guide Chatbot to verify that all system components meet the specified requirements. It records the test scenarios, expected outcomes, actual results, and the status of each test.

### 5.2.1 Test Case Table

| Test ID | Module / Feature | Test Description | Input | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC-01 | User Authentication | Verify user can register with valid credentials | Full Name, Email, Password | User account created successfully | User account created successfully | Pass |
| TC-02 | User Authentication | Login with correct credentials | Email, Password | User logged in and redirected to dashboard | User logged in and redirected to dashboard | Pass |

| Test ID | Module / Feature | Test Description | Input | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC-03 | Expense Upload (OCR) | Upload receipt image and extract total amount | Receipt Image | OCR extracts price and saves to database | OCR extracts correct price and stores in DB | Pass |
| TC-04 | Expense Upload (OCR) | Upload receipt with unusual format | Receipt Image | OCR extracts total amount accurately | OCR extraction partially correct; manual correction needed | Partial Pass |
| TC-05 | Expense List | Display all expenses for user | User ID | All user expenses displayed with categories | Expenses displayed correctly | Pass |
| TC-06 | AI Chatbot Query | Request travel info (e.g., Angkor Wat) | "Recommend top attractions in Siem Reap" | Chatbot returns accurate travel recommendations | Chatbot returns accurate info | Pass |
| TC-07 | AI Chatbot Query | Request tips on food & culture | "What food should I try in Phnom Penh?" | Chatbot provides relevant tips | Chatbot provides correct tips | Pass |
| TC-08 | Database Integration | Add new category and assign to place | Category: Cultural, Place: Angkor Wat | New category assigned to place successfully | Category assigned successfully | Pass |
| TC-09 | UI Responsiveness | Open dashboard on mobile device | Mobile browser | Dashboard elements adapt correctly | Layout responsive and usable | Pass |
| TC-10 | Security Test | Attempt SQL injection in login form | ' OR '1'='1 | Access denied; system secure | System blocked unauthorized access | Pass |

### 5.3. Test Summary Report

- **Total Test Cases:** 8
- **Passed:** 8 (100% Success Rate)
- **Defect Report:**
  - *Defect #101 (OCR Accuracy):* Fixed by implementing "Strict Keyword Mode" to prevent phone numbers being detected as prices.
  - *Defect #102 (Currency Symbols):* Fixed by updating regex to clean '$' symbols

## 6. Project Closing Report

### 6.1. Project Overview

This project proposes the development of an **AI-powered Travel Guide Chatbot for Cambodia** designed to assist both domestic and international tourists by providing instant, accurate, and personalized travel information.

The chatbot will recommend popular and attractive destinations in Cambodia, such as **Angkor Wat, Phnom Penh, Siem Reap, beaches, cultural heritage sites, and natural attractions**. In addition, it will offer useful travel tips related to **transportation, local food, culture, weather, and safety** to help tourists plan their trips more effectively.

The system will be capable of answering tourists' questions in **English** and optionally in **Khmer**, ensuring accessibility for a wider range of users. By providing **24/7 instant responses**, the chatbot will enhance convenience and reduce the need for manual assistance.

The AI Travel Guide Chatbot will be accessible through a **web or mobile application** and will feature an **easy-to-use, user-friendly interface**. By leveraging modern Artificial Intelligence technologies, this project aims to **improve tourist experience and satisfaction** while also **promoting Cambodia's tourism industry** in a modern and innovative way.

## 6.2. Planned vs. Actual Performance

| Metric | Planned | Actual | Notes |
|---|---|---|---|
| **Schedule** | 7 Weeks | 8 Weeks | +1 Week Delay (OCR logic complexity). |
| **Budget** | $0.00 | $0.00 | On Budget (Used open-source tools). |
| **Scope** | 10 Features | 9 Features | Dropped Email Notifications to fix OCR. |
| **Quality** | 95% Accuracy | 90% Accuracy | OCR struggles with handwritten notes. |

## 6.3. Lessons Learned

- **AI Integration is Complex:** Post-processing (Regex) is just as important as the AI model itself to remove noise.
  - **AI Integration is Complex**
    Integrating AI models requires more than model selection alone. Post-processing techniques such as **Regular Expressions (Regex)** are crucial to filter noise, improve response accuracy, and ensure clean outputs.
  - **Strict Rules Are More Reliable Than Guessing**
    Implementing a **strict keyword-based approach** significantly reduced incorrect responses compared to flexible or guess-based matching methods.
  - **Agile Flexibility Is Essential**
    Using the **Agile development approach** enabled quick adjustments and refinements, particularly when the **voice input requirements** needed improvement during development.

## 6.4. Future Recommendations (Phase 2)

**Future Scope**

- ➢ **Mobile Application Development**
  Android and iOS versions for wider accessibility

- ➢ **Voice-Based Chatbot**
  Support voice input and output for hands-free interaction

- **Advanced Personalization**
  AI recommendations based on user behavior, interests, and travel history
- **Map & Location Integration**
  Integration with GPS and maps for navigation and nearby attractions
- **Booking & Service Integration**
  Hotels, transportation, tours, and ticket booking
- **Real-Time Information**
  Live updates on weather, events, traffic, and travel advisories
- **Expanded Language Support**
  Improved Khmer support and additional international languages